



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001:2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2023 – 2024 (Odd Semester)

Innovative Teaching Method

UNIT I – Introduction to Optical Fiber Communication

Degree, Semester & Branch: B.E., V Semester & ECE A&B

Course Code & Title: CEC345 & Optical Communication and Networks

Name of Faculty member : Ms.L.Krishna Kumari

Name of the Topic: Phase and group velocity

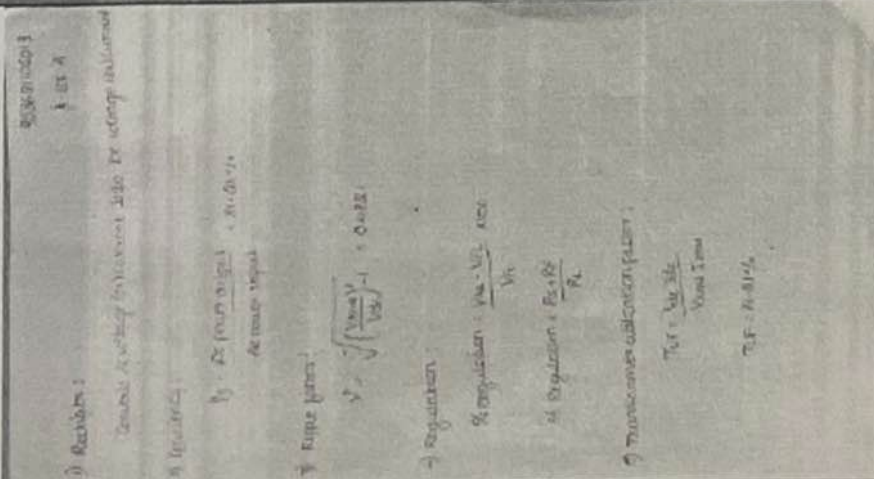
Name of the Innovative Practice: One-minute Paper

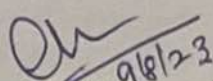
Date & Duration: 09.08.2023 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Phase and group velocity




Name of the faculty member


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UNIT II - Transmission Characteristics of Optical Fibers

Degree, Semester & Branch: B.E., V Semester & ECE A&B

Course Code & Title: CEC345 & Optical Communication and Networks

Name of Faculty member (s): Ms.L.Krishna Kumari

Name of the Topic: Intramodal Dispersion

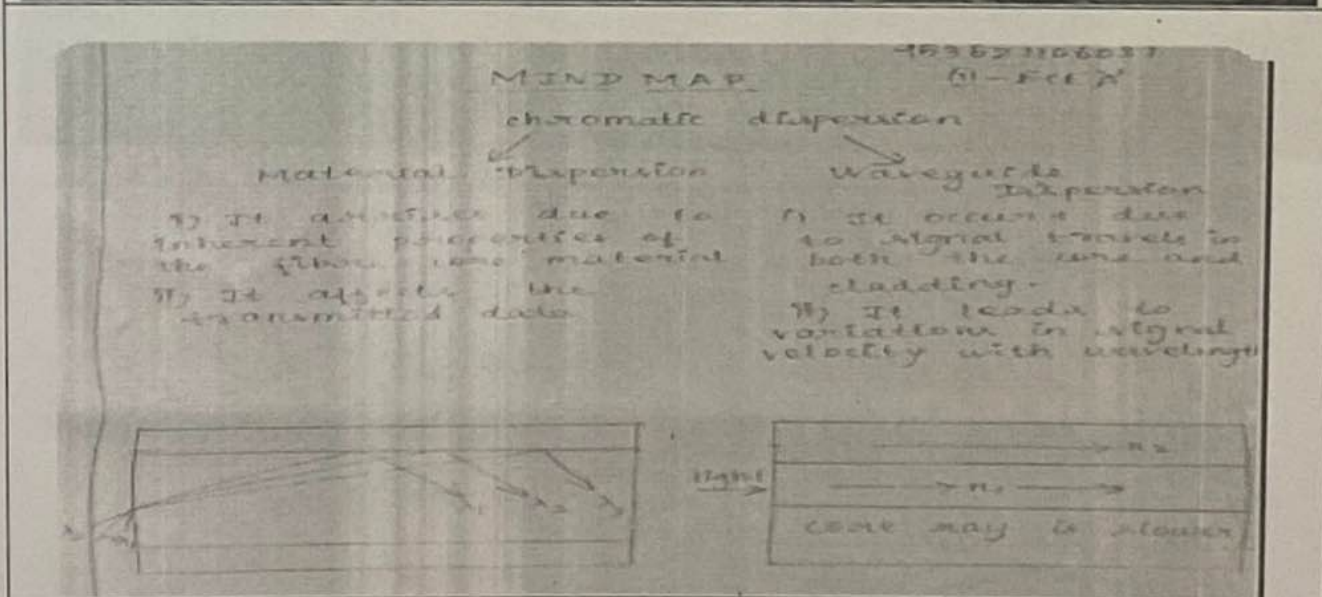
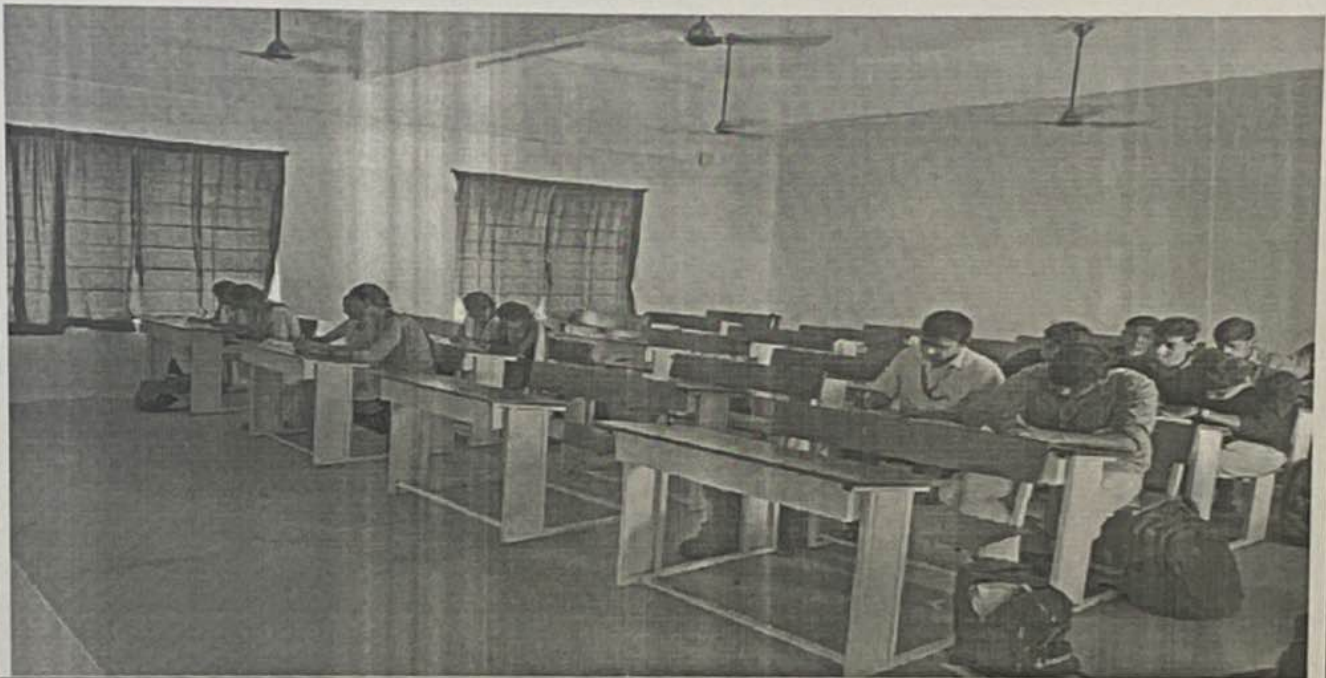
Name of the Innovative Practice: Mind Map

Date Duration: 30.08.2023 & 10 minutes

Sample Images:

Innovative Teaching Method Execution

Intramodal Dispersion



Dr. L. Krishna Kumari
20/8/23
Name of the faculty member

[Signature]
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Academic Year 2023 – 2024 (Odd Semester)

Innovative Teaching Method

UNIT III - Optical Sources and Detectors

Degree, Semester & Branch: B.E., V Semester & ECE A&B

Course Code & Title: CEC345 & Optical Communication and Networks

Name of Faculty member (s): Ms.L.Krishna Kumari

Name of the Topic: Surface and Edge emitting LED

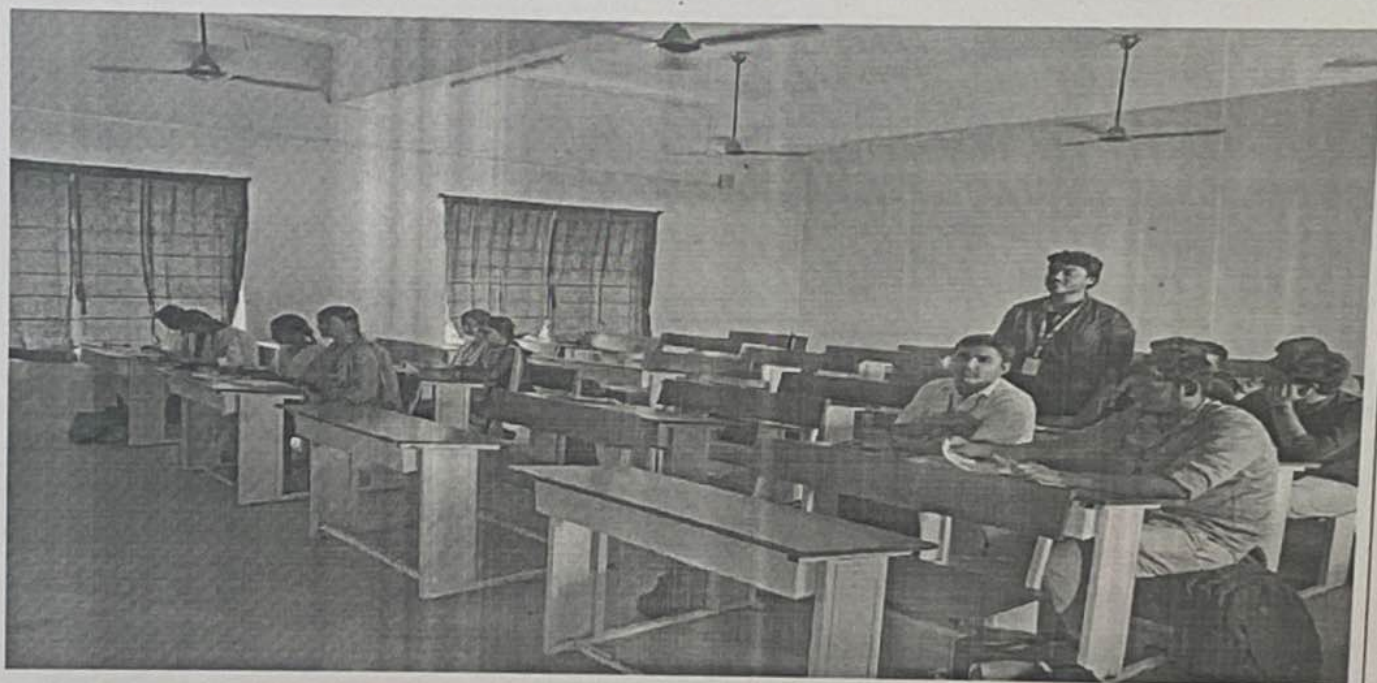
Name of the Innovative Practice: One minute speech

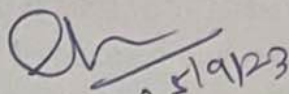
Date & Duration: 15.09.2023 & 5 minutes

Sample Images:

Innovative Teaching Method Execution

Surface and edge Emitting LED




5/9/23

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Engineering Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: B.E., V Semester & ECE A & B

Course Code & Title: CEC345 & Optical Communication and Networks

Name of Faculty member (s): Ms.L.Krishna Kumari

Innovative Practice Description

- **Unit / Topic:** IV/ Total Fiber Attenuation Measurement
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** Flipped Class room

- **Justification:**

The topic was chosen for flipped class because Total Fiber Attenuation Measurement covers attenuation analysis measurement. The Unit four deals with optical fiber parameter measurements. Since the students have to exposed to various optical fiber measurement techniques and design flipped class was given . This topic can be learnt in self-learning mode with simple manipulations and logic. The students have chance to browse the topic to know how the attenuation was calculated and its schematic representations. And find more information related to optical fiber measurement techniques. Thus this flipped class activity is chosen to make them to work out different techniques for fiber attenuation measurement.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

The overview of flipped class activity was given to students before the implementation of the activity. The students were divided into groups and one topic is allotted to each group from fiber optic measurement techniques as their wish. The topic related information and e-sources were given to the students. The students have to prepare a presentation for 10 minutes and each group has to present their topic chosen and others will listen to the presentation. After the presentation, question & answering is planned and others can share their ideas and doubts. By this way, all groups were shared their concepts learnt in their topic inside the class.

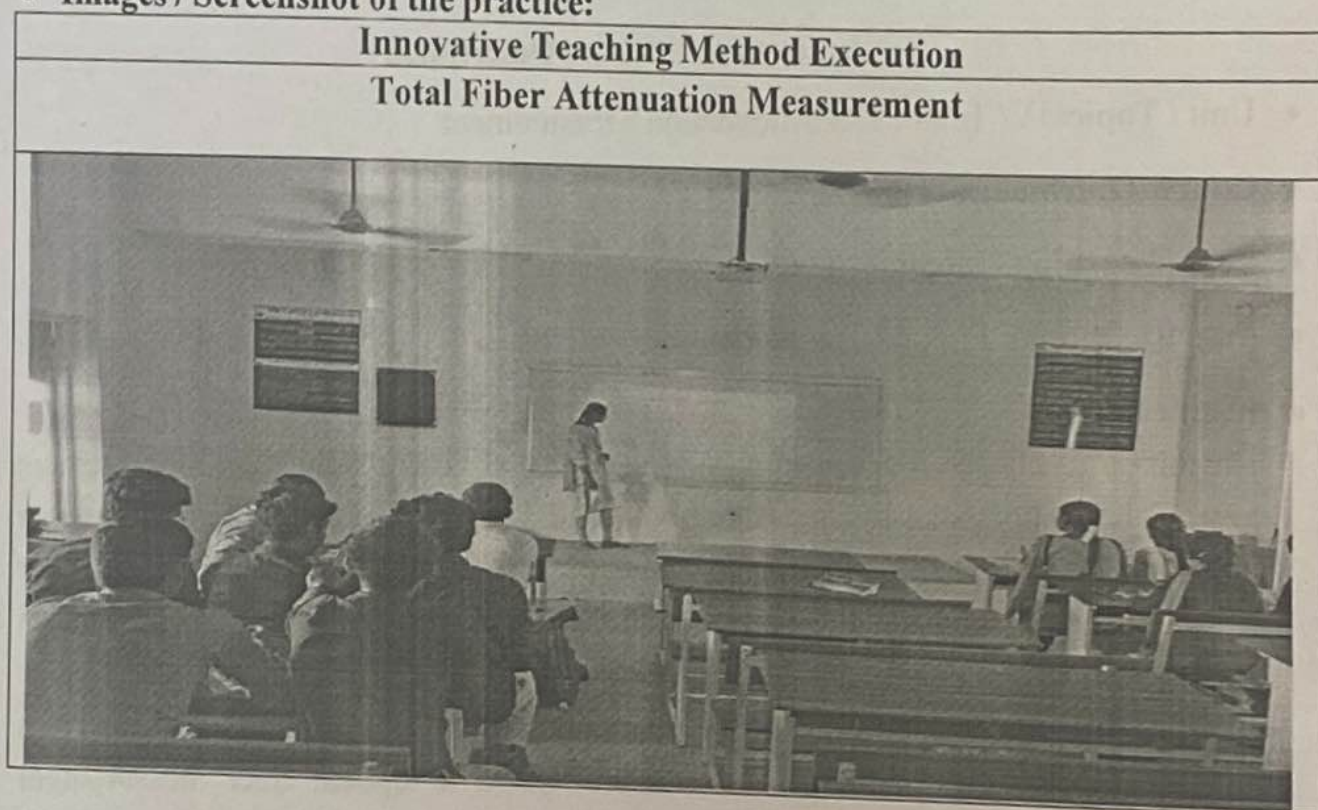
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO1	PSO3
CO4	3	3	2	2	2	1	2	2

• PO / PSO mapped

Innovative practice	PO3	PSO3
	2	2
Justification for correlation	It is mapped to level 2 because since the students will determine the error sources in digital signal transmission needs analysis of attenuation measurement.	The concept of various fiber measurement techniques are required to analyze the probability of error in optical receiver. Thus the course outcome is mapped at level 2.

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The students felt a new practice and it helped them to know to make good presentation and avoid stage fear. They liked this practice of teaching that can be done through this type of activity. They could get idea from learning sources which would help them to prepare the presentation.

❖ *Benefit of the practice:*

Students done this activity in a successful manner. They planned the activity for their chosen topic and prepared the presentation in the conceptual manner. All members in the group were learnt the topics thoroughly. By forming the group, students were able to grasp the knowledge of concept by sharing the ideas among them. Each member in the group was able to present and deliver the concept of optical fiber analysis and design.

❖ *Challenges faced in implementation:*

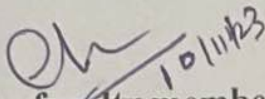
Some of the students felt difficult to present because of lack of communication. Few groups done their presentation beyond their time limit. Few students in the group felt difficult to answer all the questions.

Textbooks:

1. John M.Senior, "Optical Fiber Communication", Pearson Education, Fourth Edition.2010.

Other References:

1. Gred Keiser, "Optical Fiber Communication", McGraw Hill Education (India) Private Limited. Fifth Edition, Reprint 2013.
2. Govind P. Agrawal, "Fiber-Optic Communication Systems", Third Edition, John Wiley & Sons, 2004.
3. J.Gower, "Optical Communication System", Prentice Hall Of India, 2001
4. Rajiv Ramaswami, "Optical Networks " , Second Edition, Elsevier , 2004.
5. P Chakrabarti, "Optical Fiber Communication", McGraw Hill Education (India)Private Limited, 2016


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Innovative Practices Description

UNIT V - Optical Communication Systems and Networks

Degree, Semester & Branch: V Semester B.E.ECE A & B

Course Code & Title: CEC345 – Optical Communication and Networks

Name of the Faculty member: Ms.L.Krishna Kumari

Name of the Topic: Optical Switching Networks

Name of the Innovative Practice: Think pair share

Date & Duration: 25.10.2023 & 30 Minutes

ICT Tool Used: Smart Class Room

Description:

Think-pair-share (TPS) is a collaborative learning strategy where students work together to solve a problem or answer a question about an assigned reading. This strategy requires students to think individually about a topic or answer to a question; and share ideas with classmates. Discussing with a partner maximizes participation, focuses attention and engages students in comprehending the reading material.

Goals (Learning Outcomes):

- The students will be able to Describe the various types of Optical Switching Networks.

Use of appropriate methods:

Justification for choosing the topic using Think pair share activity:

Since the students are asked to think on the particular topic, they are able to think, remember and discuss the topics with their friends in the classroom. They are able to share their ideas on the particular topic by having discussions with them.

Effective presentation Implementation (Plan & Execution) with Proof:

The students were in the class room. The topic was explained to the students first. Then they were asked to think about the topic and after sometime they were asked to pair themselves and to have a discussion on the particular topic.

Innovative Teaching Method Execution

Think pair share on "Optical Switching Networks"



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when the students were able to answer to the question on the particular topic when asked on the next week.
- ✓ The students understood the topic very clearly since they had discussion with their classmates.

Reflective Critique:

• Benefits:

The students were more interested in learning the concepts if they think and discuss the topic and ask questions. They clarified their doubts with the neighboring students and the faculty. They learnt the concepts of various categories of Optical Switching network architecture by think pair share activity. They asked more doubts in the topic and learnt many things.

• Challenges:

Basically some time is needed to explain the basic concepts of switching networks. The slow learners found somewhat difficult in the initial stage later they also were able to think and share the concepts with their friends.

CO – PO / PSO mapping:

CO	PO 1	PO 2	PO 3	PO 4	PO 9	PO 10	PO 12	PSO 1	PSO 3
CO5	3	3	3	3	2	2	1	2	2

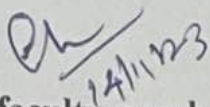
CO 5. Examine the fundamentals of optical network transmission modes, layers, protocols and optical network deployment techniques.

Textbooks:

1. John M.Senior, "Optical Fiber Communication", Pearson Education, Fourth Edition.2010.

Other References:

1. Gred Keiser, "Optical Fiber Communication", McGraw Hill Education (India) Private Limited. Fifth Edition, Reprint 2013.
2. Govind P. Agrawal, "Fiber-Optic Communication Systems", Third Edition, John Wiley & Sons, 2004.
3. J.Gower, "Optical Communication System", Prentice Hall Of India, 2001
4. Rajiv Ramaswami, "Optical Networks ", Second Edition, Elsevier , 2004.
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